RESEARCH · 02

Not All Goals Are *Equal*.

A TIME-AWARE EXPECTED-POINTS MODEL · TOP 5 LEAGUES + EREDIVISIE

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renball

TL;DR

Three *sentences*

THE MODEL

Price every goal by the state it creates minus the state it ends, from 2,377,080 team-minutes in 13,206 matches.

THE CLOCK

A go-ahead goal from level is worth 1.51 EP in the last 15, 2.05× the first 15 (95% CI 1.94–2.18). Level late, the upside is 1.58× the downside.

THE PLAYERS

No "clutch" signal: only 17 of 1,090 players scored at clearly better-than-average times, fewer than chance.

THE QUESTION

Every goal counts as *one*. Should it?

An early opener, a late winner and a consolation at 4–0 down all add one to the scoreboard.

The first nudges the odds. The second nearly settles them. The third changes almost nothing. A goal's worth depends on the scoreline it changes and on how much time is left to change it back.

So price it: the expected final points of the scoring team just after the goal, minus just before it. A late equaliser is worth 0.96 EP; a late second goal at 1–0 only 0.31.

DATA

Six leagues, seven *seasons*

39,403	14,360	13,206	2,377,080
GOALS PRICED	MATCHES LOADED	MATCHES IN THE MODEL	TEAM-MINUTES

FBref match pages, 2017–2018 to 2023–2024: Premier League, La Liga, Serie A, Bundesliga, Ligue 1, Eredivisie. Shot tables, official scores, per-match minutes and ages.

Own goals are missing from the shot table: in 1,154 matches (8.0%) the rebuilt score falls short of the official one. They are left out of the value model.

307 of 2,068 Eredivisie matches have no usable data.

METHOD

Two tables, one *subtraction*

STATE VALUE

What a scoreline is worth

Every minute a team spent at a goal difference in a 15-minute window, averaged over the points those teams finished with (3 · win + 1 · draw). 0–0s and goalless spells count.

GOAL VALUE

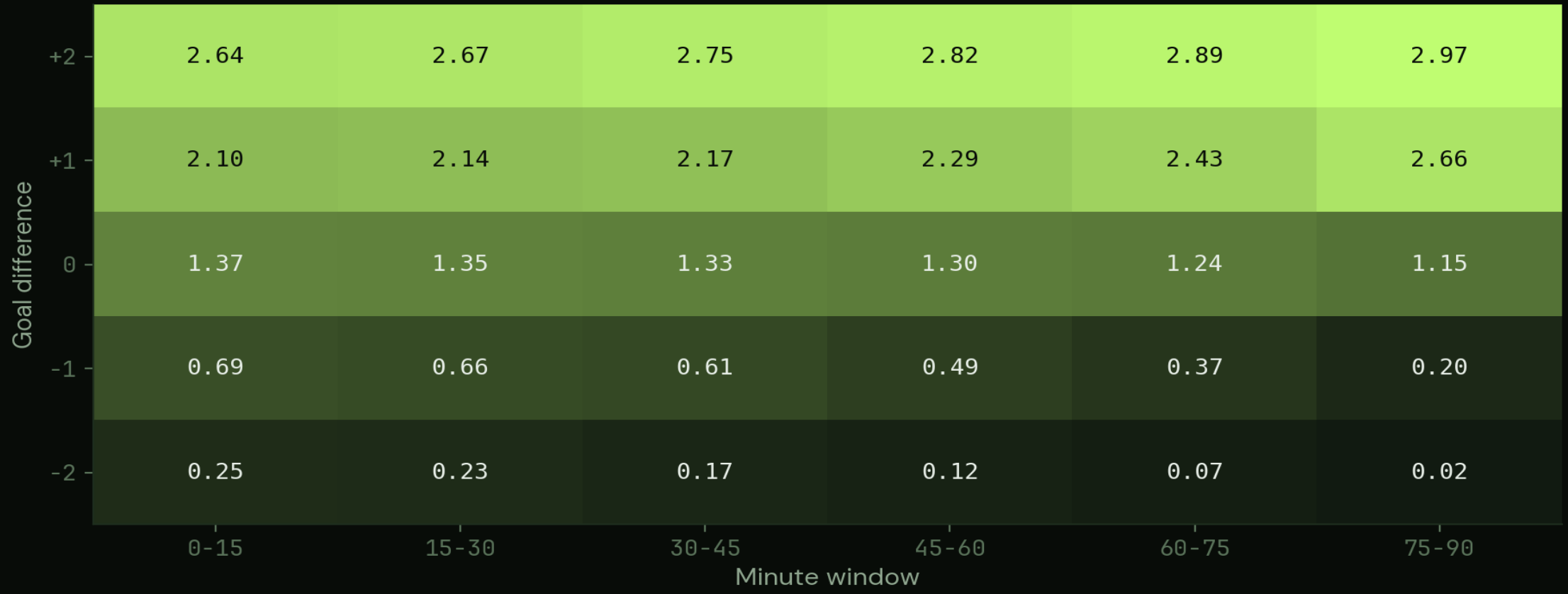
What a goal is worth

The state the goal creates minus the state it ends, in the same window. Time-blind version: the same with all minutes pooled.

Intervals: 1,000 bootstrap resamples of whole matches within each league · fixed seed · cells seen in fewer than 30 matches flagged sparse

STATE VALUES

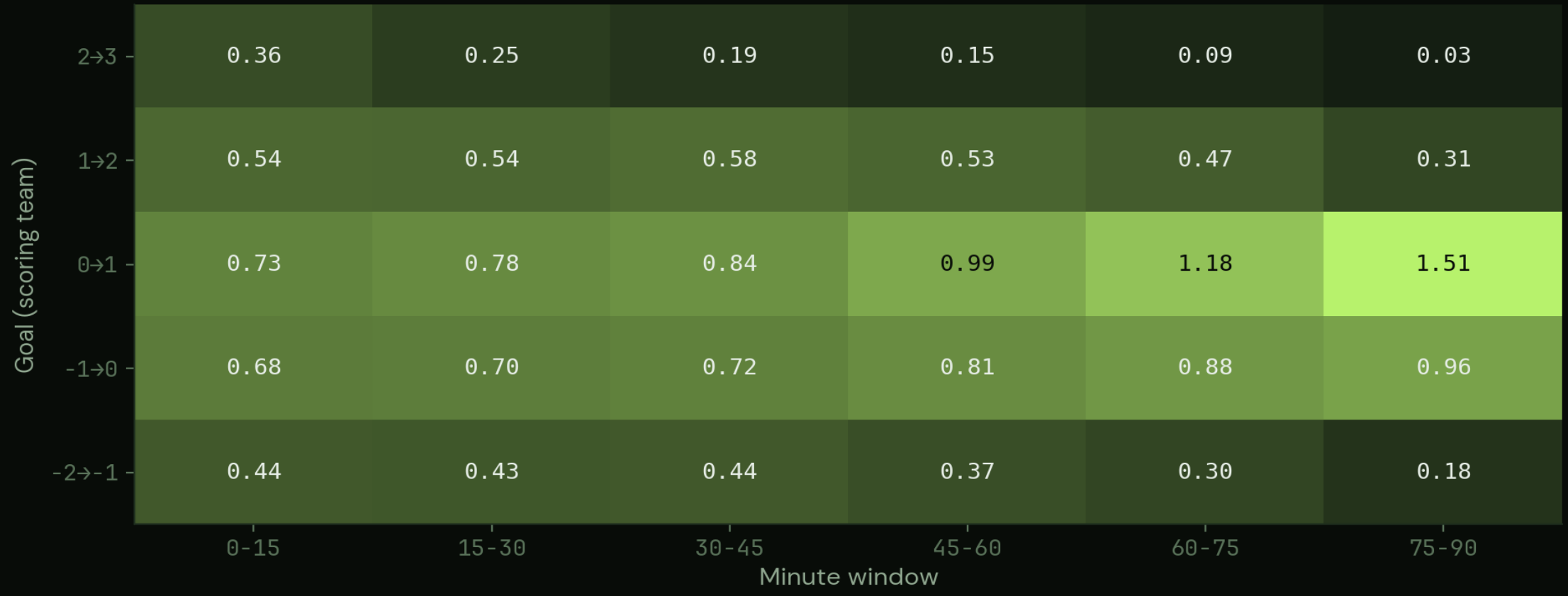
What each scoreline is *worth*



Expected final points by goal difference and window. One up: 2.10 early, 2.66 late. Level: 1.37 → 1.15.

GOAL VALUES

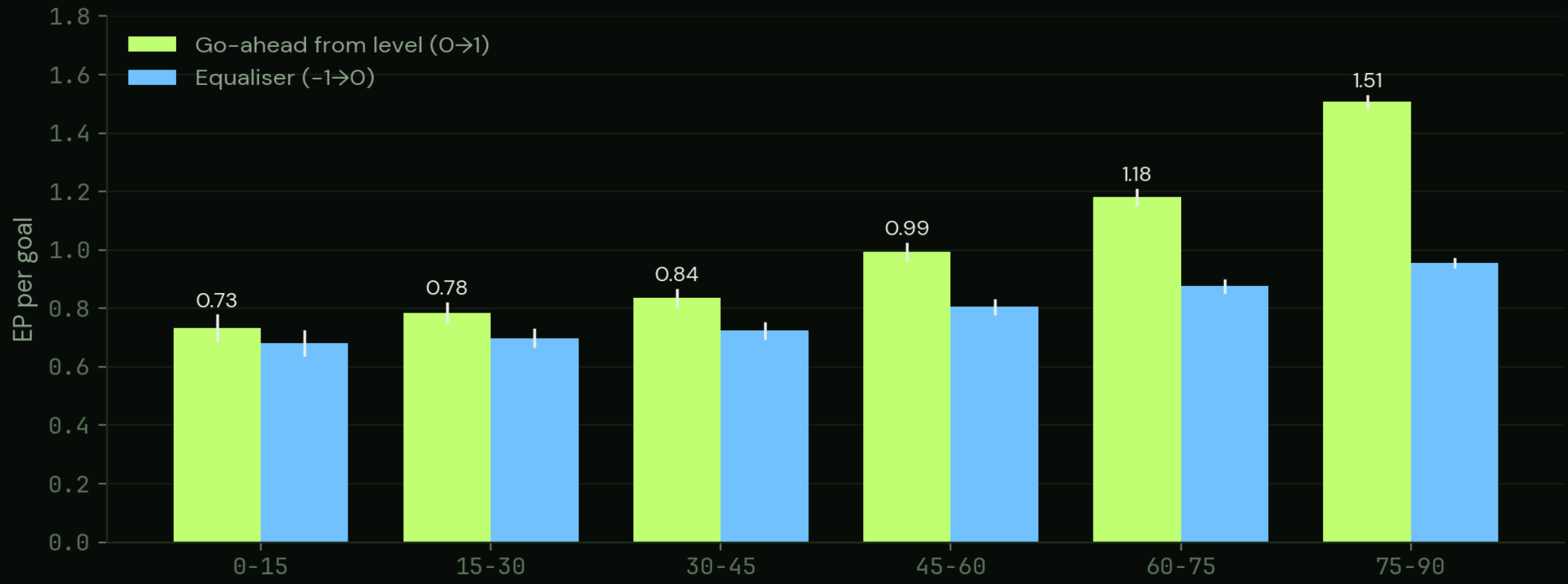
The value of a goal, by *minute*



State after minus state before. Brightest: 0→1 in the last 15 (1.51 EP, 2,113 goals). Leads stretched late are worth little.

HEADLINE

The opener: *2.05×* late vs early



0→1: 0.73 EP in the first 15, 1.51 in the last 15 (ratio 95% CI 1.94–2.18). Late equaliser 0.96 EP. Whiskers: 95% intervals.

GAME MANAGEMENT

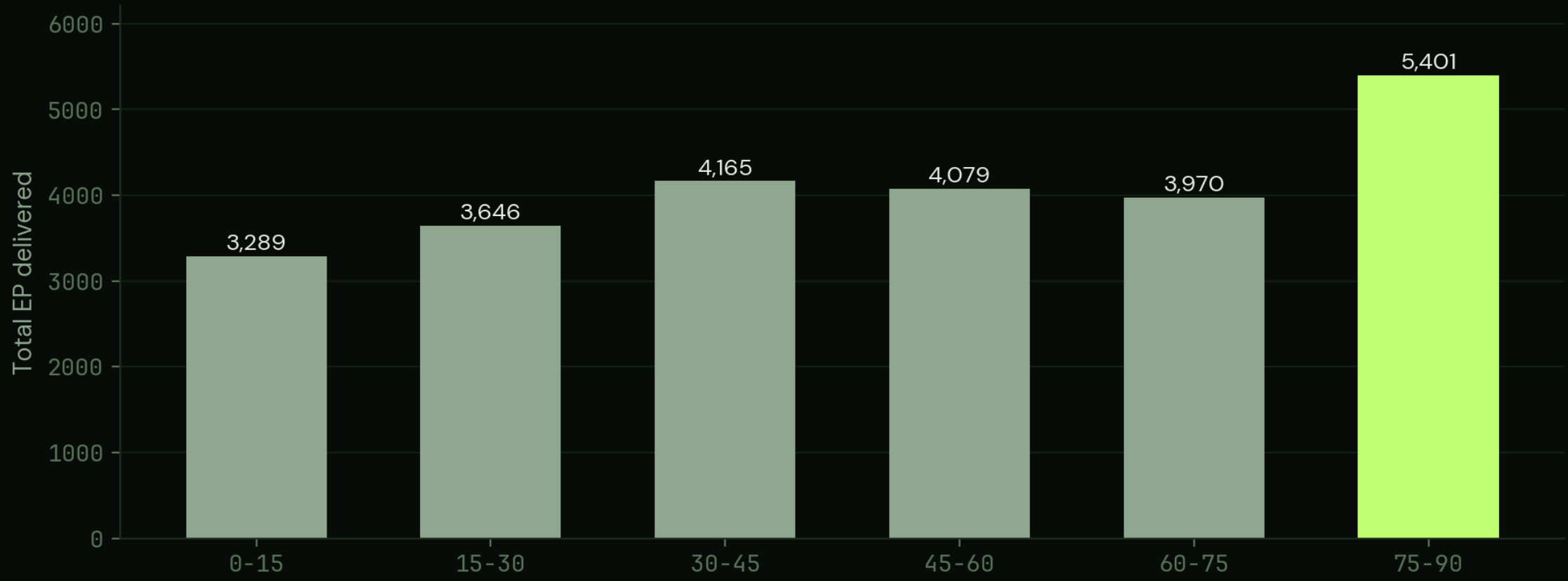
Next goal: *upside ÷ downside*

SCORE	0-15	15-30	30-45	45-60	60-75	75-90
Leading by one	0.74×	0.68×	0.69×	0.53×	0.40×	0.20×
for / against (EP)	0.54 / 0.73	0.54 / 0.78	0.58 / 0.84	0.53 / 0.99	0.47 / 1.18	0.31 / 1.51
Level	1.08×	1.12×	1.16×	1.23×	1.35×	1.58×
for / against (EP)	0.73 / 0.68	0.78 / 0.70	0.84 / 0.72	0.99 / 0.81	1.18 / 0.88	1.51 / 0.96
Trailing by one	1.56×	1.63×	1.65×	2.15×	2.90×	5.39×
for / against (EP)	0.68 / 0.44	0.70 / 0.43	0.72 / 0.44	0.81 / 0.37	0.88 / 0.30	0.96 / 0.18

Value of scoring the next goal ÷ cost of conceding it. Level in the last 15: a goal for is worth 1.51 EP, a goal against costs 0.96: 1.58× (95% CI 1.56–1.60). Prices, not a test of tactics.

BY WINDOW

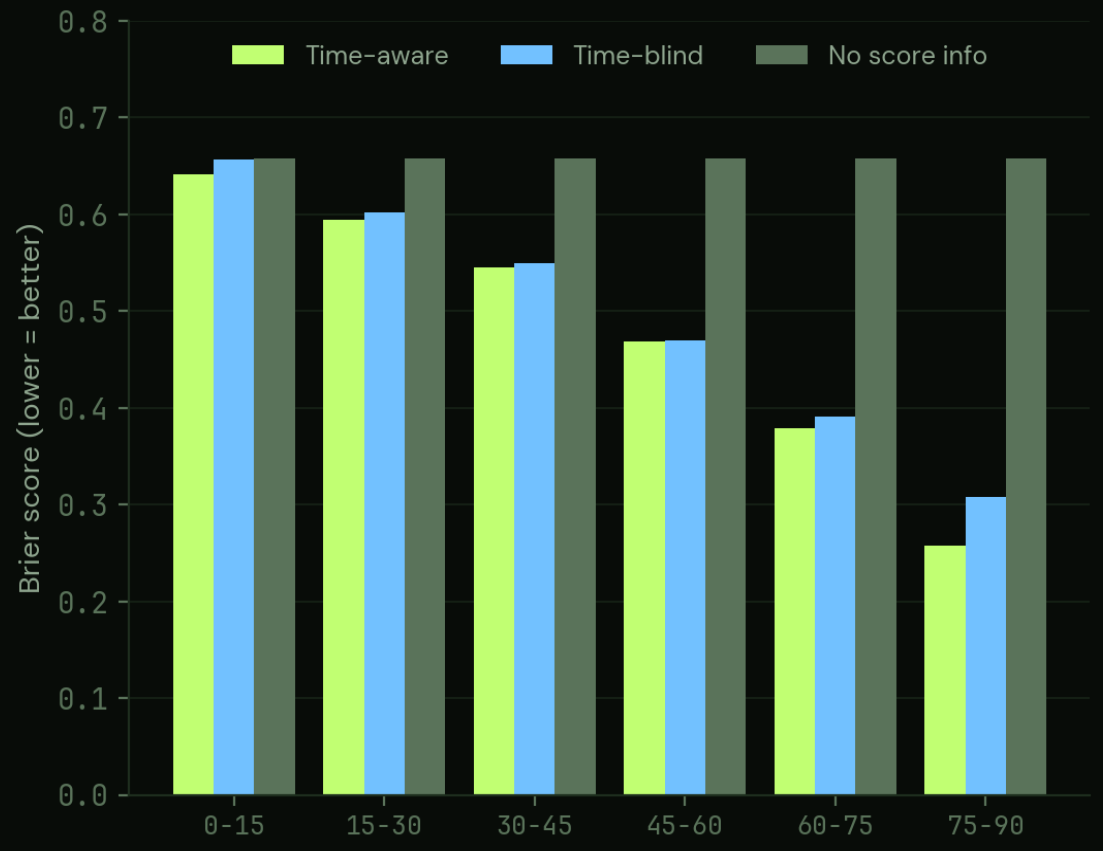
Where the points get *cashed in*

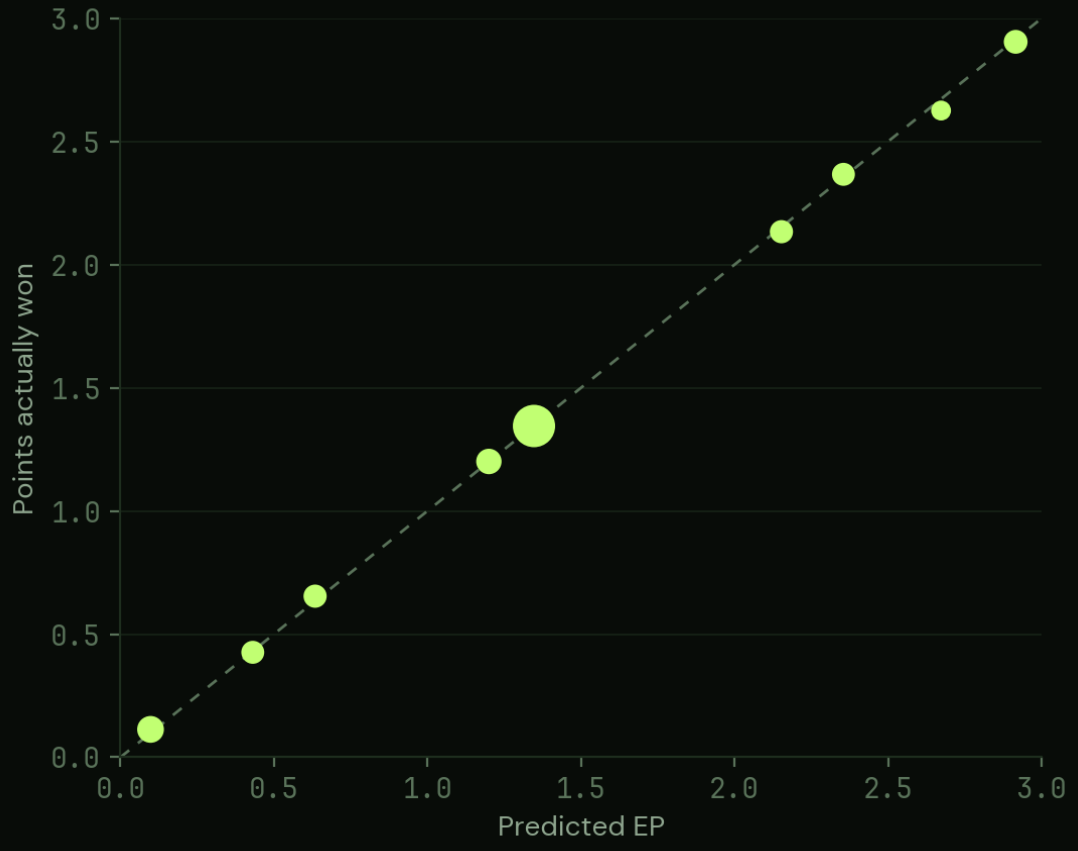


Every goal at its own value. Last 15 vs first 15: 1.64× (95% CI 1.54–1.76): 8,238 goals against 4,596.

OUT OF SAMPLE

It predicts seasons it never *saw*





Fit on 2017-2018 to 2021-2022; predict 3,838 matches in 2022-2023 and 2023-2024. Brier 0.481 vs 0.496 time-blind; largest calibration gap 0.05 EP.

ROBUSTNESS

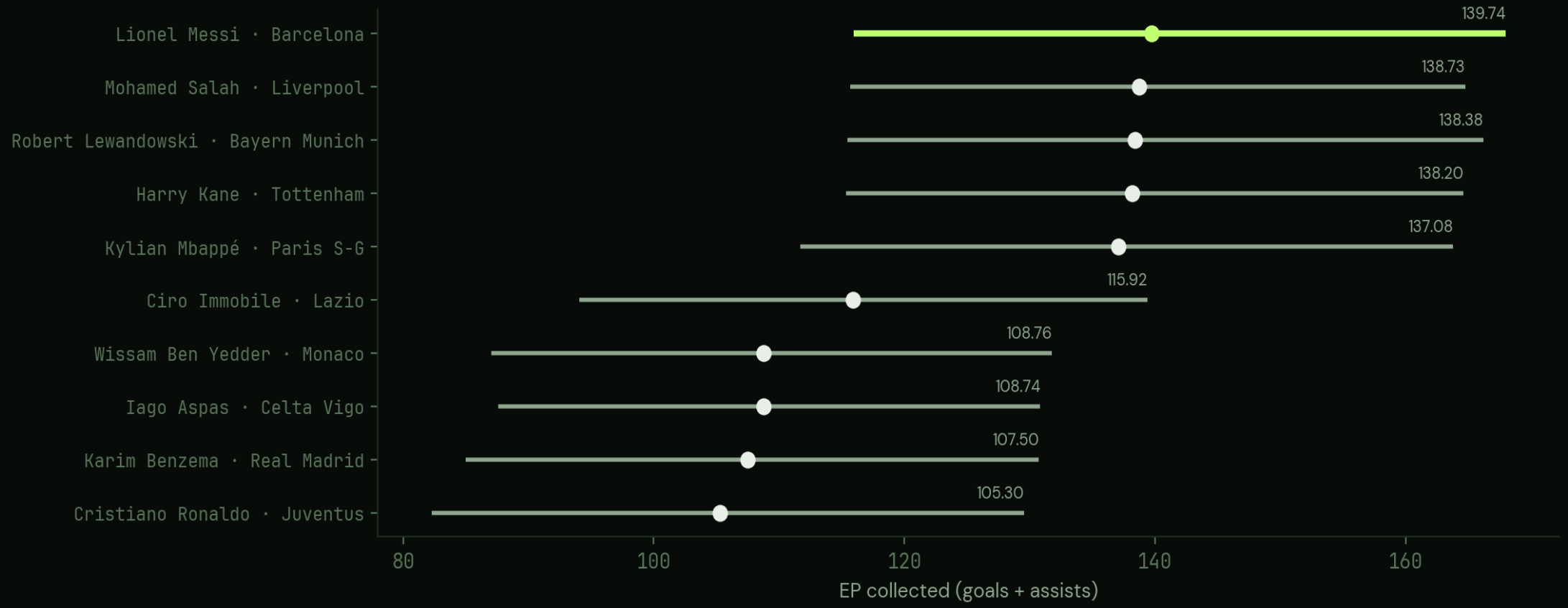
Does it *hold up*?

VERSION	MATCHES	OPENER LATE	OPENER EARLY	RATIO	95% CI
Primary model	**13,206**	**1.51**	**0.73**	**2.05×**	**1.94–2.18**
Own-goal matches put back	14,360	1.44	0.71	2.03×	n/a
Fit on 2017–18 to 2021–22	9,368	1.52	0.75	2.04×	n/a
Fit on 2022–23 to 2023–24	3,838	1.46	0.70	2.09×	n/a
Premier League	2,407	1.52	0.72	2.11×	1.82–2.47
La Liga	2,476	1.52	0.66	2.31×	1.98–2.77
Serie A	2,440	1.49	0.79	1.90×	1.67–2.21
Bundesliga	1,972	1.50	0.81	1.85×	1.65–2.12
Ligue 1	2,286	1.53	0.72	2.11×	1.85–2.47
Eredivisie	1,625	1.48	0.70	2.11×	1.78–2.59

Late vs early go-ahead goal from level. League range 1.85× to 2.31×; league differences inside it not tested.

EP COLLECTED

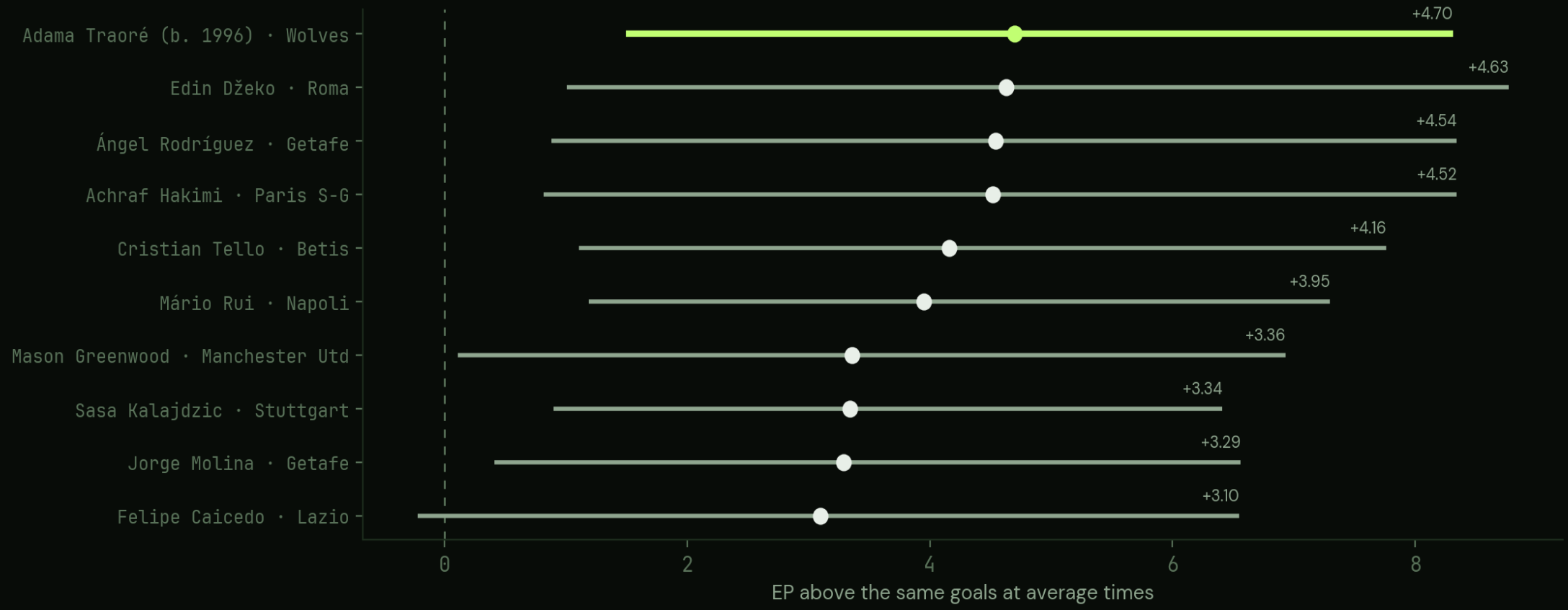
Top 10: a *statistical tie* at the top



Top five within 2.66 EP, intervals overlapping. Goals + SCA1, once per player per goal. Bars: 95% intervals.

TIMING PREMIUM

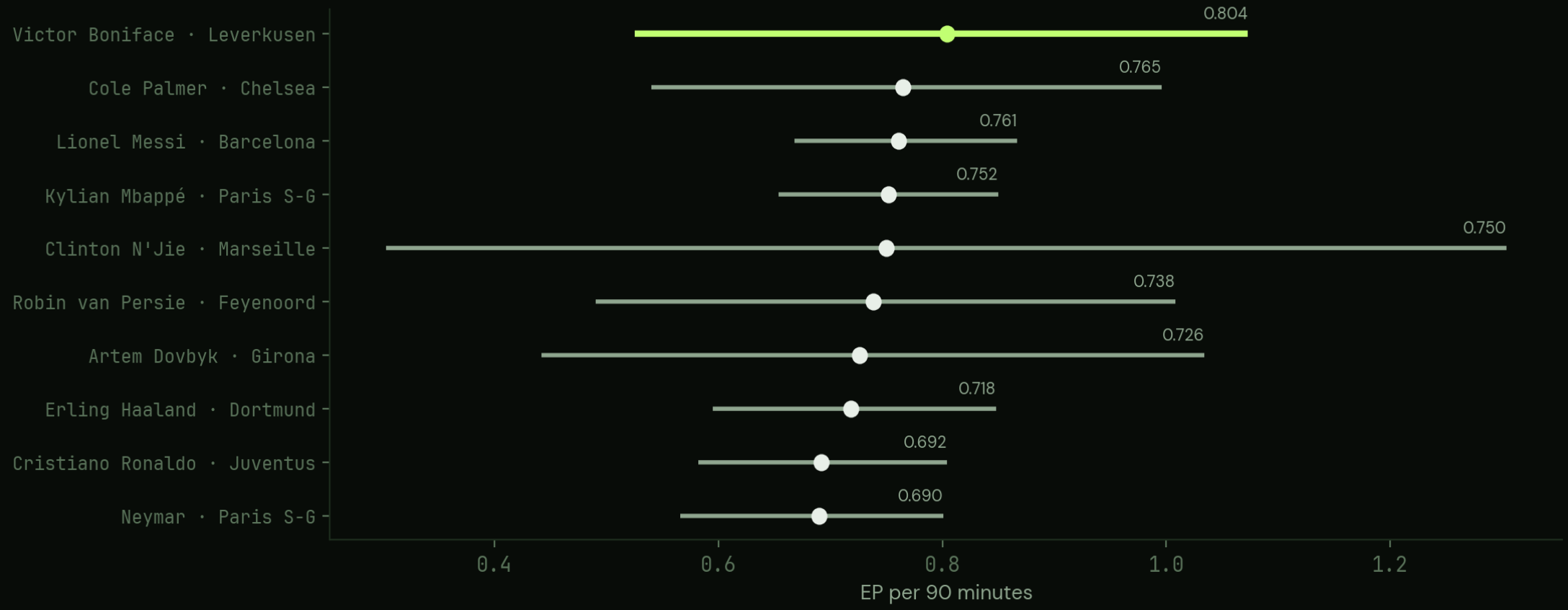
Clutch? Not beyond *chance*



Only 17 of 1,090 players (20+ involvements) clearly above zero; chance alone gives ~27.

RATE

EP per 90: hot streaks *float up*



At least 900 minutes. 4 of the top ten played under 3,000 minutes; their intervals are the widest.

TAKEAWAYS

Who should *care*

Coaches

TRAINING & TACTICS

EXPLORATORY

The last 15 minutes deliver 1.64× the expected points of the first 15: more goals, higher stakes.

36,201 goals · 95% CI 1.54–1.76× · training implication untested

Managers

MATCH-DAY DECISIONS

MODERATE

Level after 75': a goal for is worth 1.51 EP, a goal against costs 0.96. The upside is 1.58× the downside.

95% CI 1.56–1.60× · all 6 leagues · prices outcomes, not tactics

MODERATE

1–0 up after 75': conceding costs 1.51 EP, a second goal adds only 0.31.

ratio 95% CI 0.19–0.22 · values, not the effect of sitting deeper

Clubs

RECRUITMENT & CONTRACTS

MODERATE

Don't pay for 'clutch': only 17 of 1,090 players scored at clearly better-than-average times, fewer than chance (~27).

20+ goal involvements · bootstrap intervals · repeatability not tested

Agents

POSITIONING A PLAYER

No defensible claim for this audience from this study.

EVIDENCE STRONG MODERATE EXPLORATORY Definitions: methodology § Evidence labels

LIMITATIONS

What this *can't* tell you

- Prices, not causes: a goal's value is measured on what teams went on to do, not isolated from red cards or subs.
- No team strength: a 1–0 lead is priced the same for favourite and underdog.
- Own goals: 1,154 matches left out of the model; including them moves the headline from 2.05× to 2.03×.
- Every goal counts as one (no xG); "assist" is FBref's SCA1; 15-minute windows.
- Timing premiums and per-90 rates are not tested for repeatability.

CONCLUSION

What this *changes*

The clock changes the price. *Clutch doesn't.*

The same go-ahead goal from level is worth 2.05× as much in the last 15 minutes as in the first, in every league and in seasons the model never saw. That makes the table usable for game management: level late, a goal for is worth 1.58× what a goal against costs. It doesn't single out clutch players: compared with the same scorelines at average times, almost nobody's timing stands out from chance.

Thanks for reading.

Full methodology, source code and data:

renball.com

```
/PROJECTS/02-POINTS-BY-GAMESTATE/

→ methodology      the model, the bootstrap, the held-out test, limitations

→ dashboard        every player, state and goal value by league and season

→ code/            matches.py · value_model.py · build.py
```